



ST BENEDICT'S COLLEGE

SOURCE BOOKLET
LIFE SCIENCES
2025 YEAR-END EXAMINATION PAPER II
GRADE 11

QUESTION 1

Read the information below on human evolution. Use this information as well as your own knowledge to answer Question 1 in the question paper.

In the first half of the 20th century, the region thirty miles northwest of Johannesburg, South Africa, produced so many fossils of our early ancestors that it became known as the Cradle of Humankind. It was declared a World Heritage Site in 1999. The Sterkfontein Caves in this area contain the discovery of a 2,3-million-year-old fossil *Australopithecus africanus* (nicknamed "Mrs. Ples"). Sterkfontein alone has produced more than a third of early hominid fossils ever found prior to 2010. Nearby this site, the Rising Star Cave system contains the Dinaledi Chamber (chamber of stars) in which were discovered fifteen fossil skeletons of an extinct species of hominid, named *Homo naledi*. The Dinaledi Chamber contains over 1 500 *H. naledi* fossils, the most extensive discovery of a single hominid species ever found in Africa.

Hominid fossils discovered by Lee Berger and his team of researchers

Lee Berger, an American paleoanthropologist started working at the University of the Witwatersrand in the early 1990s and began to hunt for hominid fossils. But the spotlight in human evolution had long since shifted to the Great Rift Valley of East Africa. Most researchers regarded South Africa as an interesting side story to human evolution but not the main plot. Berger was determined to prove them wrong.

What he most wanted to find were fossils that could shed light on the main outstanding mystery in human evolution; the origin of our genus, *Homo*, between two million and three million years ago. On the one side are the apelike australopithecines, such as *Australopithecus afarensis* (3,85 – 2,95 mya) and its most famous representative, Lucy, a skeleton discovered in Ethiopia in 1974. On the other side is *Homo erectus* (2 mya – 100 000 years ago), a tool-wielding, fire-making, globetrotting species of the "out of Africa" hypothesis with its big brain and body proportions much like ours. Within that almost million-year gap between them, a bipedal animal was transformed into a self-aware human being, *Homo sapiens*, a creature not just adapted to its environment but able to apply its mind to master it. How did that evolution happen?

The fossil record is frustratingly unclear. Slightly older than *H. erectus* is a species called *Homo habilis*, or "handy man" – because it is believed to be responsible for the stone tools that were found at Olduvai Gorge in Tanzania. Berger has long argued that *H. habilis* was too primitive to deserve its privileged position at the root of our genus. Some other scientists agree that it really should be called *Australopithecus*. But Berger has been nearly alone in arguing that South Africa was the place to look for the true earliest *Homo*. Then, in 2008, he made a truly important discovery. In Malapa, he and his nine-year-old son, Matthew, found some hominid fossils. Dated to about 2 million years ago, they were the first major finds from South Africa in decades. In most ways, they were very primitive, but they had some oddly modern traits too. Berger decided the skeletons were a new species of australopithecine, which he named *Australopithecus sediba*.

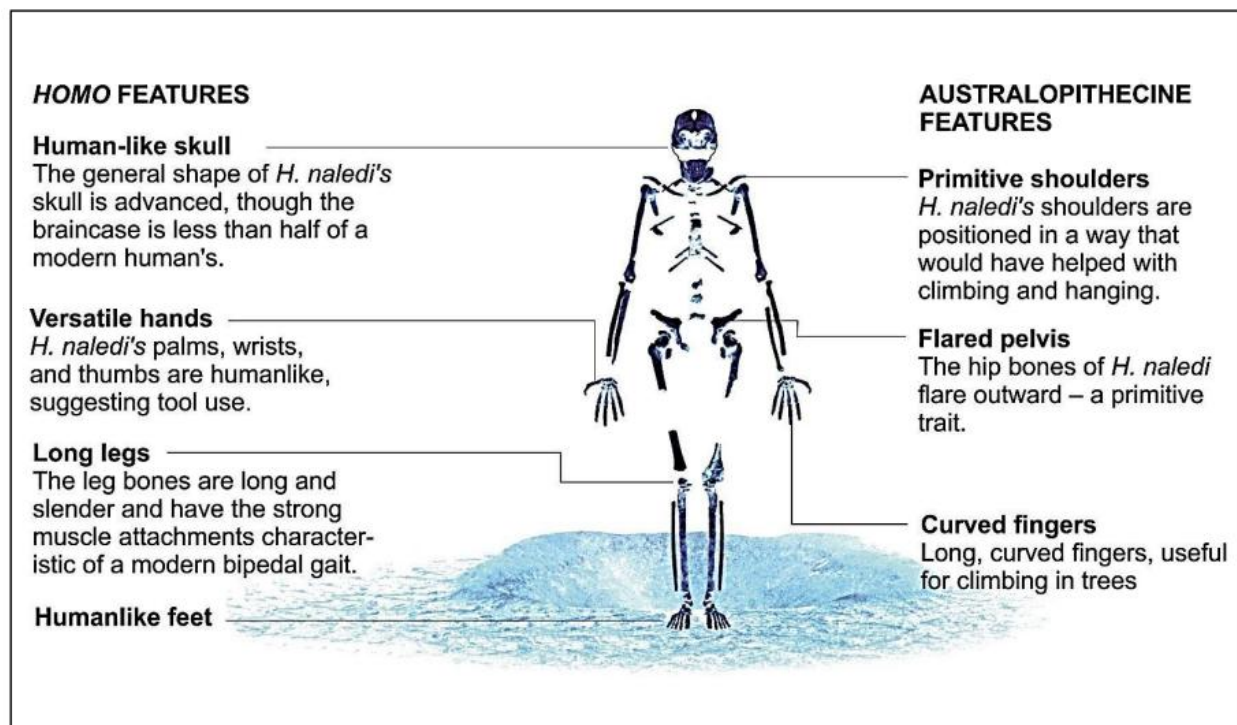
Then in 2012, some cavers discovered bones in the Rising Star Cave system. Parts of the skeletons looked astonishingly modern. But other parts were just as astonishingly primitive – in some cases, even more apelike than the australopithecines

Global collaboration in analysing the Rising Star fossils

In palaeoanthropology, specimens are traditionally kept confidential until they can be carefully analysed and the results published, with full access to them only given to the discoverer's closest collaborators. This would mean that answering the mystery of the Rising Star find could take years, even decades. Berger wanted answers by the end of the year. In his opinion all researchers in the field of palaeontology should have access to important new information as quickly as possible.

Along with about 20 senior scientists who had helped him work on *A. sediba*, Berger invited more than 30 young scientists to Johannesburg from about 15 countries, for an intensive "fossil fest" lasting six weeks. To some older scientists who weren't involved, putting young people on the front line just to rush the research seemed hasty. But for the young scientists in question, it was "a paleo fantasy come true. At university, you dream of a pile of fossils no one has seen before, and you get to figure it out." Their findings were astounding; "You could almost draw a line through the hips – primitive above, modern below. The message we're getting is of an animal right on the cusp of the transition from *Australopithecus* to *Homo*," said Berger.

Image 1: Summary of important features of *Homo naledi*.

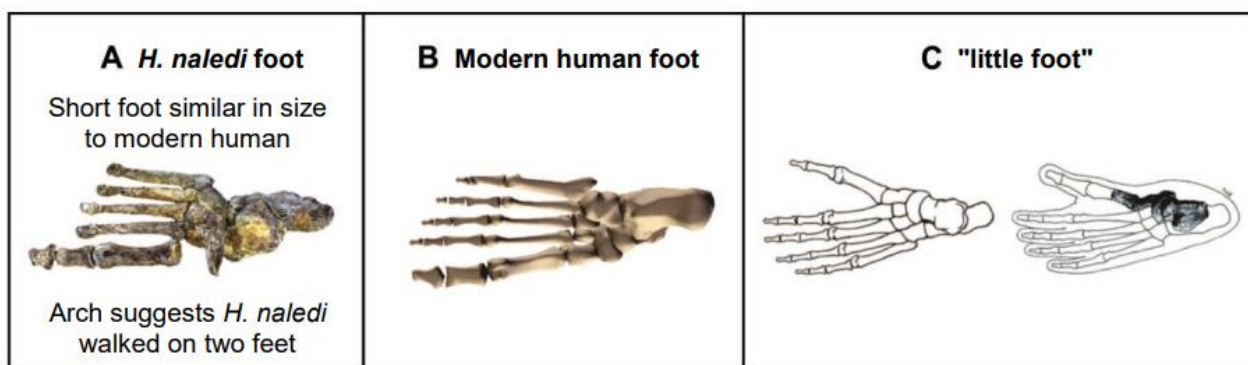


[Adapted from: <https://news.nationalgeographic.com>]

Two important features of *Homo naledi*

1. The general structure of the head looks advanced enough to be called *Homo* but their braincases were tiny – a mere 560 cubic centimetres (cc/cm³) for the males and 465 cubic centimetres for the females. This is far less than *H. erectus*'s average of 900 cubic centimetres, and well under half the size of our own (*Homo sapiens*). A large brain is the key to humanness, representative of a species that has evolved to live by its wits.
2. The feet were virtually indistinguishable from our own. Except for the slightly curved toes, the arches suggest an efficient long-distance stride.

Image 2: Diagrams showing the feet of A- *Homo naledi*; B- modern human; C- Little Foot



[Adapted from: <https://www.researchgate.net>]

To add to the mystery of their origin, mixed soil sediments in the cave where *H. naledi* was found make it difficult to date the bones. Berger and his team believe that *H. naledi* was even closer to modern humans than *Homo erectus* is, and it clearly belonged in the *Homo* genus. They called it *Homo naledi*, named for the cave where the bones had been found. Berger also hypothesises that *H. naledi* disposed of their dead in one of the chambers of the caves where the bones were found. Disposal of the dead is a feature of humanity. It brings closure for the living, confers respect on the departed, or assists in their transition to the next life. But *H. naledi*, Berger emphatically stresses, was not human – which makes the behaviour all the more intriguing.

[Adapted from: <https://news.nationalgeographic.com> and <https://researchgate.net>]

QUESTION 2

Read the information below on the evolution of hominid skulls. Use this information as well as your own knowledge to answer Question 2 in the question paper.

Slicing meat may have aided human evolution

The change in diet of hominids from a plant-based diet to a more meat-eating one meant less energy was required for chewing. Sliced meats, usually cooked, are part of many modern diets. But our ancestors may have been eating raw slices. And that could have led to changes in the human body, a new study finds. Early member of *Homo*, the human genus, had a flair for preparing raw sliced meat, a new study suggests. That meaty diet may have literally changed the face of *Homo* evolution. It also may have enable advances in talking and walking. The early human ancestor that used stone tools to slice up raw meat before eating it is known as *Homo erectus*. These early hominids lived some 1,8 million years ago in Africa and Eurasia. Slicing meat would have made it possible to consume more calories while reducing chewing effort, say Katherine Zink and Daniel Lieberman. They are paleoanthropologists at Harvard University. They reported their findings online on March 9, 2017 in Nature.

With less chewing needed, faces and jaws could get smaller, they conclude. And an energy-rich diet allowed hominid brains and bodies to enlarge. The shrinkage of bones and muscles involved in chewing also would have affected other parts of the body, these scientists argue. An expanded vocal tract would have boosted the ability of these early ancestors to make speech sounds. And a realignment of the base of the skull would have led to a repositioning of the spinal cord. That led to an increased ability to walk and run long distances. *Homo erectus* (shown below) and other early members of the *Homo* genus evolved relatively small jaws and faces because they sliced raw meat with stone tools. That would have made it easier to chew and digest, the study concludes.

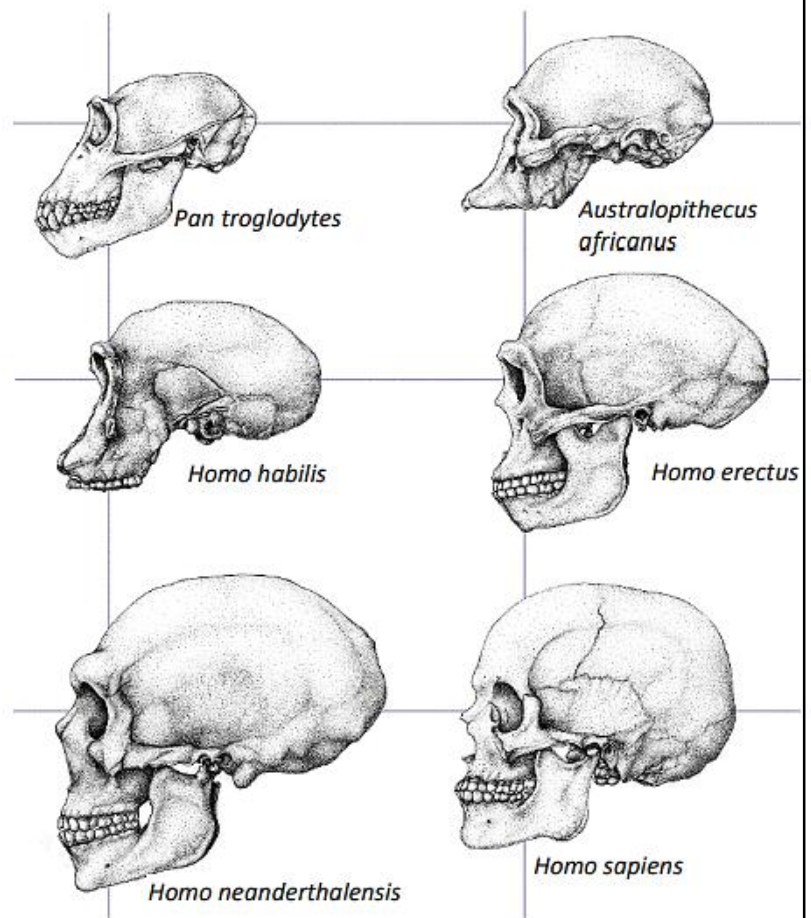


Image 3: diagram of skulls of hominids and a chimpanzee

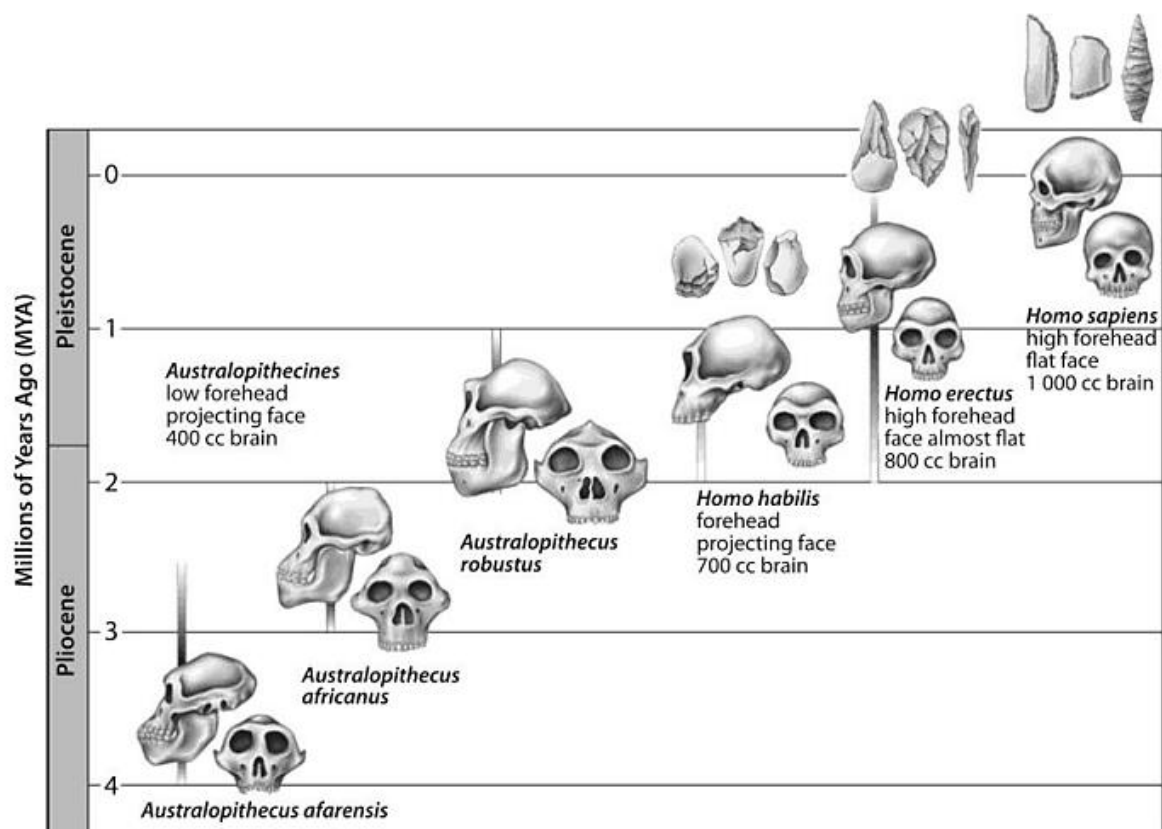
Zink and Lieberman made their findings through a study of modern-day humans. A total of 34 adults were given measured portions of goat meat or three plants rich in starch. These were jewel yams, carrots and red beetroots. Food came either unprepared, sliced, cooked, or pounded using a stone to soften it. The volunteers had one job: Chew the food. The scientists then analysed how much the face muscles worked and tallied the number of chews needed to ready the different foods for swallowing. After an average of 40 chews, participants still struggled to break apart 3-gram chunks of raw goat meat. Cooked chunks were no easier to chew into pieces. Meat sliced from raw, 3-gram chunks, however, required an average of about 31 chews to break into pieces small enough to easily swallow and digest. The researchers assumed that *H. erectus* may have gotten one-third of its total calories from sliced meat. The rest would have come from stone-pounded plants, such as jewel yams. With that ratio, *H. erectus* would have needed to chew its food 17 percent less and exert 26 percent less force than if only unprepared plants were consumed, the scientists now estimate. Over a year, they calculate, dining on sliced meat could have saved *H. erectus* more than 2,5 million chews.

Image 4: Images of the lower jaw and teeth of a chimpanzee (left), Australopithecus afarensis (middle) and Homo sapiens (right)



[Adapted from: <https://www.quora.com/Why-don't-humans-have-chimpanzees-like-teeth>]

Cooking makes it even easier to chew and digest meat and edible plants such as yams, Zink and Lieberman noted. But they estimate that our hominid ancestors began to regularly cook food only about 500 000 years ago. Cooking, the researchers say, would have stimulated further jaw and facial shrinkage in *Homo sapiens* – our species – but not in *H. erectus*. Instead, they conclude, a taste for sliced, raw meat got those facial changes off to a fast start in *H. erectus*.

Image 5: Graph showing the increase in brain size of hominids over time

[Adapted from: <https://www.mhhe.com/biosci/genbio/>]

Researchers have long noted that *H. erectus* skulls display relatively smaller jaws and faces than relatives that preceded the *Homo* genus more than 2 million years ago. The new study provides the first evidence for a decades-old assumption that tools for meat-eaters prompted the evolution of smaller faces in early *Homo*, says Manuel Domínguez-Rodrigo. He works at the Complutense University of Madrid in Spain. "The key is the consumption of sliced meat," he says. And it, he notes, was "enabled by the use of stone tools. *H. erectus* must have regularly hunted prey to maintain a diet of about one-third meat," Domínguez-Rodrigo adds.

[Adapted from: <https://www.sciencenewsforstudents.org/article/slicingmeatmayhaveaidedhumanevolution>]

QUESTION 3

Read the information below. Use this information as well as your own knowledge to answer Question 3 in the question paper.

SOURCE A

Neanderthals (*Homo neanderthalensis*) are our closest extinct relatives. They lived in Europe between 400 000 and roughly 40 000 years ago. It is likely that they evolved from a population of *Homo erectus* in Europe.

Some defining features of their skulls include the large middle part of the face, angled cheek bones, and a huge nose for adding moisture and warming cold, dry air. Their bodies were shorter than humans; another adaptation to living in cold environments, but their brains were often larger than ours.

Neanderthals made and used a diverse set of sophisticated tools, controlled fire, lived in shelters, made and wore clothing, and were skilled hunters of large animals. They occasionally made symbolic or ornamental objects. There is also evidence that Neanderthals deliberately buried their dead and occasionally even marked their graves with offerings, such as flowers. No other primates, and no earlier human species, had ever practised this sophisticated and symbolic behaviour.

The reason for their extinction is still debated.

[Adapted from: <https://humanorigins.si.edu>]



Artist's impression of a Neanderthal

[Adapted: <http://secure.i.telegraph.co.uk>]



Human skull

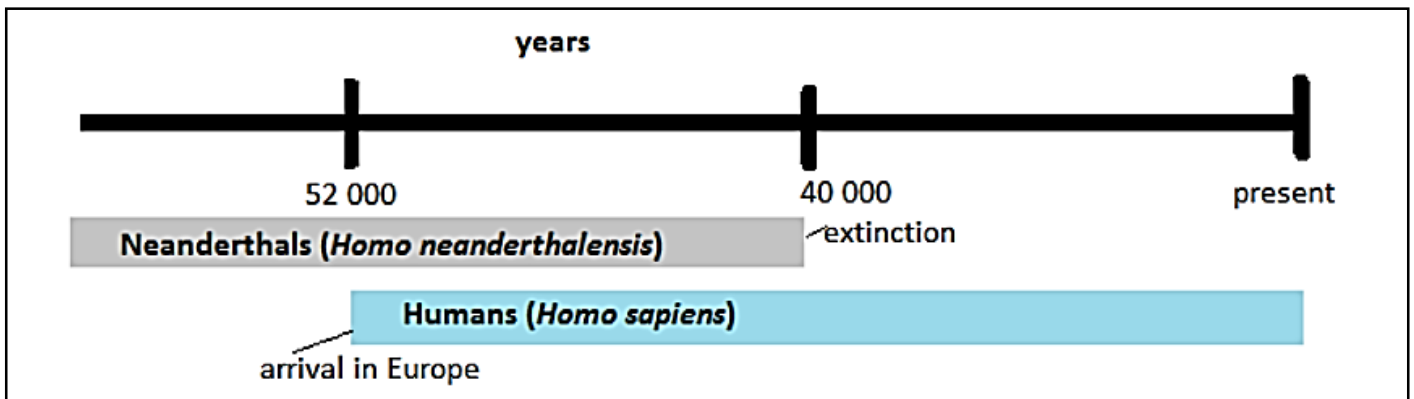


Neanderthal skull

[Adapted: <http://pulic.media.smithsonianmag.com>]

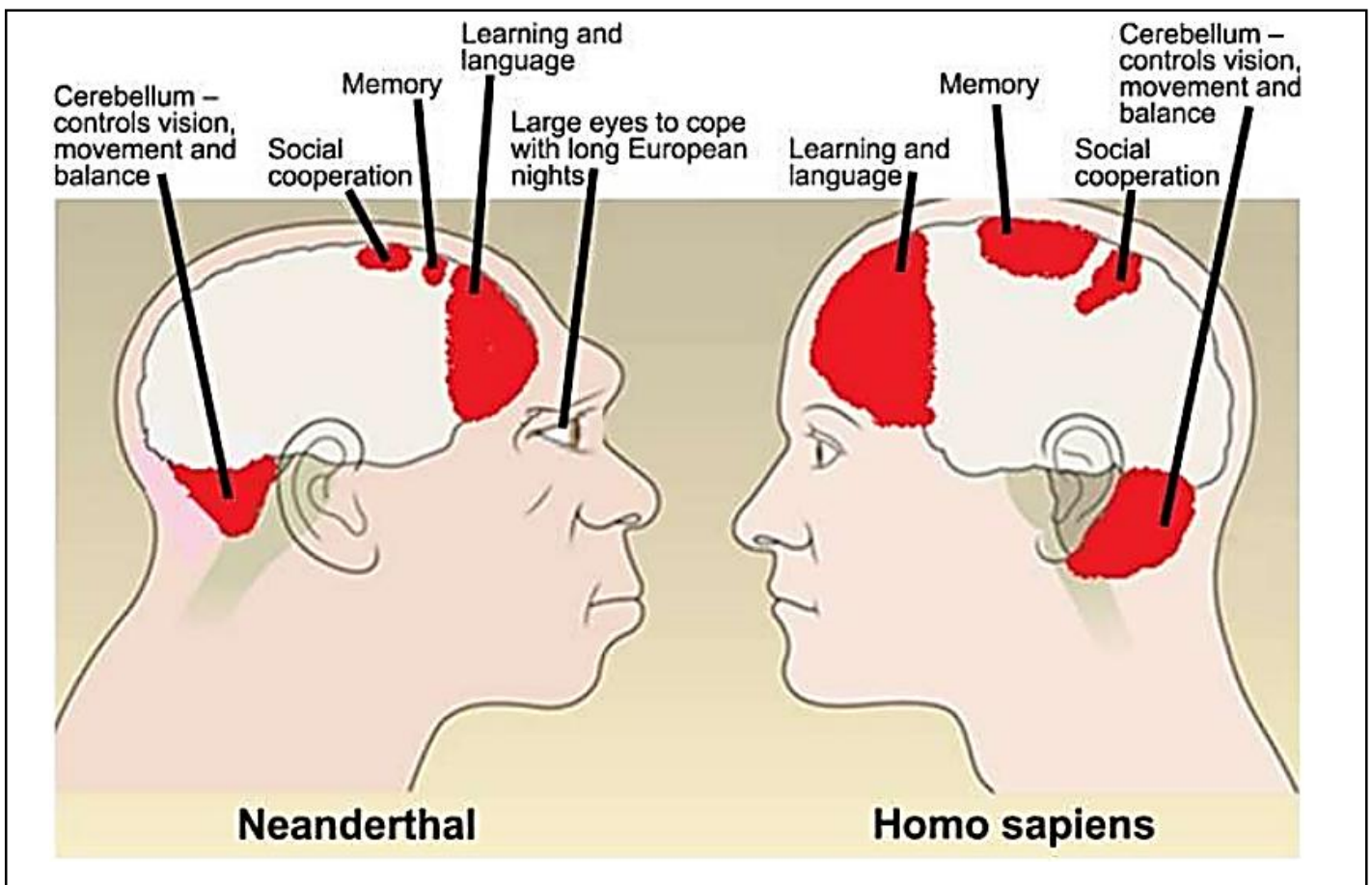
The most significant wave of emigration of *Homo sapiens* out of Africa took place about 70 000 years ago, spreading rapidly along the coast of Asia and reaching Australia by around 65 000– 50 000 years ago. *H. sapiens* reached Europe 52 000 years ago and met the Neanderthals.

[Adapted: Rito, T. et al. 2019. A dispersal of *Homo sapiens* from southern to eastern Africa immediately preceded the out-of-Africa migration. *Sci. Reports* 9(4728)]



[Adapted: Cambridge University www.cam.ac.uk]

SOURCE B BRAIN DIFFERENCES



[Adapted: <https://i.warosu.org>; <How Neanderthals Got Their Unusually Large Brains. Livescience>]

SOURCE C INTERBREEDING***Homo sapiens* caused Neanderthal extinction**

A detailed comparison of Neanderthal and modern human genomes has revealed evidence that humans and Neanderthals interbred. Between 0% and 2,6% of DNA in humans comes from Neanderthals. People from Europe share the most DNA with Neanderthals. Does this mean that Neanderthals "disappeared" due to interbreeding with humans till they no longer existed as a distinct group?

Various fossils of hybrid individuals (individuals with one Neanderthal parent and one human parent) exist as well, including the "child of Lagar Velho", a skeleton found at Lagar Velho in Portugal



[Adapted: <https://blogs.scientificamerican.com>; Paul, M. & Short, R. 2011. Neanderthal-human Hybrids]

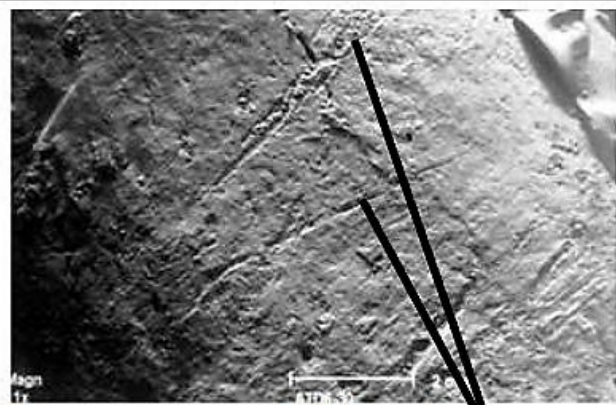
Interbreeding between Neanderthals and humans can only account for a certain degree of Neanderthal population decrease.

[Adapted: Jones, D. 2007. The Neanderthal within. New Scientist: 193(2593: 28–32)]

SOURCE D CONFLICT

Some authors have discussed the possibility that Neanderthal extinction occurred due to violent conflict with *Homo sapiens*. These conflicts involved competition over limited resources, such as prey and water.

There are markings that have been found on both human and Neanderthal bones that have probably come from weapons that were used by both groups at the time.



[Adapted: <www.neanderthals.com>]

Markings

Image: weapon markings on Neanderthal bones

SOURCE E HUMAN DISEASES

Pathogens and parasites carried by *Homo sapiens* could have been spread among the Neanderthal population. These included pathogens causing diseases such as tuberculosis, typhoid, whooping cough, encephalitis and common colds. Neanderthals would have had limited immunity to these diseases as they had not been exposed to them previously.

[Adapted: Underdown, S. 2015. Brookes research finds modern humans gave fatal disease to Neanderthals. Oxford Brookes University news]

SOURCE F SOCIAL PROBLEMS AND HUNTING

Genetic studies show that Neanderthals lived in small inbred isolated groups. These studies also show that the genetic variation present in Neanderthals was very small.

In 2006, two anthropologists of the University of Arizona also proposed that Neanderthal division of labour within these small groupings was less developed than in the larger *Homo sapiens* groups. This resulted in less efficient use of resources from the environment as compared to *Homo sapiens*.

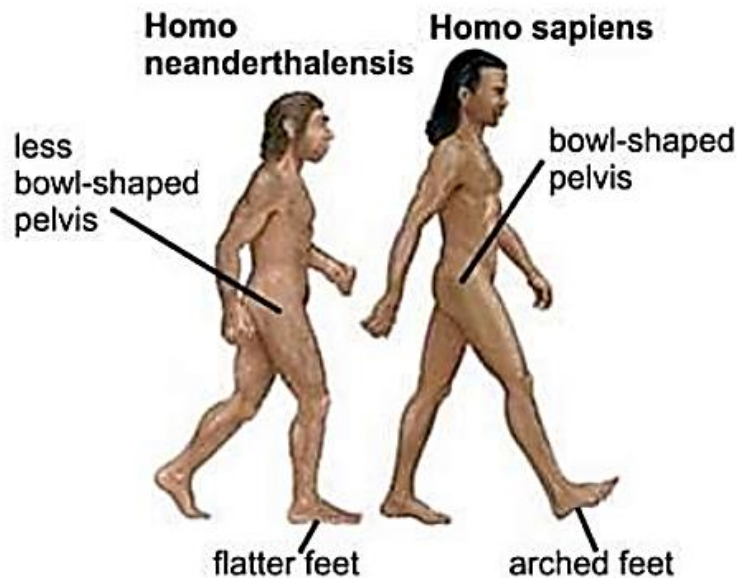
The fact that Neanderthal groups were isolated also meant that there was less cooperation between the groups. This would have been a huge disadvantage in the colder climate of Europe. When times got hard for one group, help could not be sought from another.

Studies of the teeth of Neanderthal children indicate that they grew much more quickly than modern human children. This indicates that Neanderthal children were dependent on their parents for a shorter time than human children. Neanderthal children must have had much less opportunity to learn important things like hunting and other survival skills from their parents and other group members.

[Adapted: <https://www.theguardian.com/science>]

[Adapted: <https://news.nationalgeographic.com>]

Yoel Rak, from Tel Aviv University in Israel, has noted that the fossil records show that Neanderthal pelvises in comparison to modern human pelvises would have made it much harder for Neanderthals to absorb shock and to bounce off from one step to the next, giving modern humans another advantage over Neanderthals when running and walking. Humans were therefore more able to compete with Neanderthals when hunting.



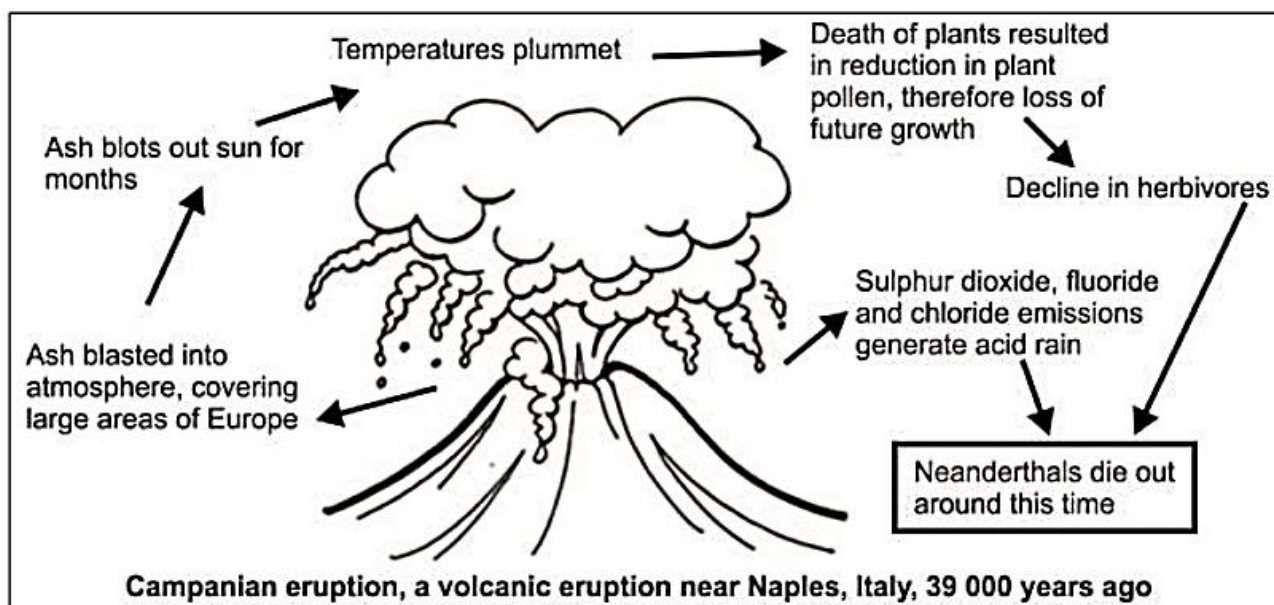
Domestication of the dog gave modern humans an advantage when hunting. They could use their dogs to help them to hunt more effectively. The oldest remains of domesticated dogs were found in Belgium (31 700 years ago) and in Siberia (33 000 years ago). Humans therefore could hunt more efficiently than Neanderthals.

[Adapted: Kuhn, S.L. & Stiner, M.C. 2006. What's a Mother to Do? The Division of Labor among Neanderthals and Modern Humans in Eurasia. *Current Anthropology* 47(6): 953-955]

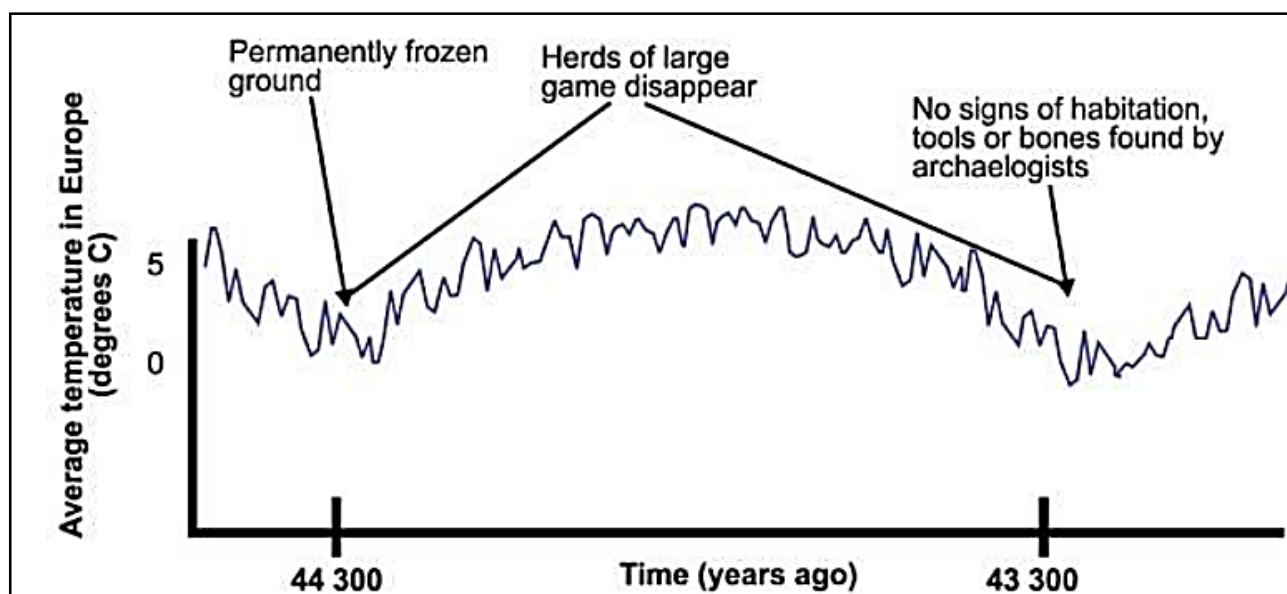
[Adapted: Weaver, T.D. & Hublin, J.J. 2009. Neanderthal birth canal shape and the evolution of human childbirth *Proc. Nat. Acad. Sci.* 106(20): 8151-8156]

SOURCE G

VOLCANIC ERUPTIONS AND CLIMATE CHANGE



We didn't kill the Neanderthals; climate change did, new study says



Homo sapiens may have had an advantage during times of climate change thanks to them having a more varied diet, which included a larger share of plants and fish than that of the Neanderthals, who relied more heavily on the large mammals that would have been most affected by the change in climate. Tel Aviv University archaeologists suggested that Neanderthals had evolved their large chests and pelvises precisely to better metabolise the high amounts of protein they were getting from their diet. However, this high level of specialisation may have contributed to their doom when conditions changed and there were fewer animals around to hunt.

[Adapted: Fish, R.V., Giovanni, O, Ort, M.& Heiken, G.1993. Mobility of a large-volume pyroclastic flow—emplacement of the Campanian ignimbrite, Italy. *Journal of Volcanology & Geothermal Research* 56(3):205–220]

SOURCE H DIET, HUNTING AND ART**Did starving Neanderthals eat each other?**

Neanderthals lived a desperately tough life, sometimes so close to starvation that when one of them died, their compatriots would devour it. This meant that they were always vulnerable to extinction.

[Rosas, A. 2006. Did starving Neanderthals eat each other? *Proceedings of the National Academy of Sciences* 10:1073]

Did Neanderthals die out because of the paleo diet?

The cold climate of Europe meant that Neanderthal diets comprised of very few fruits and vegetables. Their food consisted mostly of animal meat— fat and protein. Although fat is easier to digest, it's scarce in cold conditions, as prey animals themselves burn up their fat stores and grow lean. So Neanderthals must have eaten a great deal of protein, which is tough to metabolise and puts heavy demands on the liver and kidneys to remove toxic by-products. Neanderthals therefore had a very narrow diet. If any of these prey animals disappeared, they would have lacked food and would be vulnerable to extinction. *Homo sapiens* had technology that Neanderthals didn't, including sewing needles to make clothing in cold conditions. *Homo sapiens* also had innovative tools like bows and arrows to hunt for food.

[Adapted: <https://www.smithsonianmag.com>]

Did the Neanderthals' poor art ability contribute to their extinction?

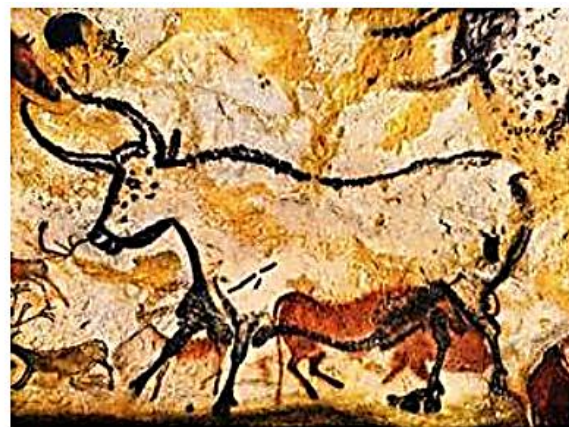
Analysis of charcoal drawings and engravings by primitive *Homo sapiens* reveals elaborate depictions of animals including lions and deer in a variety of scenes. Of Neanderthal art, however, there is virtually no evidence, and what disputed examples there are, mainly comprise simple straight lines and no recognisable images.

Scientists at the University of California have said this is closely related to a lack of hand-eye coordination due to the fact they lived off easy-to-kill prey.

Since the act of drawing enhances observational skills, drawings are useful for planning future hunts, selecting vulnerable body areas as targets and causing group cohesiveness via spiritual ceremonies. Neanderthals could not do this.



Neanderthal cave painting, Spain, 60 000 years old



Homo sapiens cave painting, Lascaux Cave, France, 40 000 years old